

Supply, Demand, and Price:

The Theory

Chapter 3

Demand

- Demand: The willingness and ability of buyers to purchase different quantities of a good at different prices during a specific period of time.
- The Law of Demand: As the price of a good rises, quantity demanded of that good falls; as the price of a good falls, quantity demanded of that good rises, *ceteris paribus*.

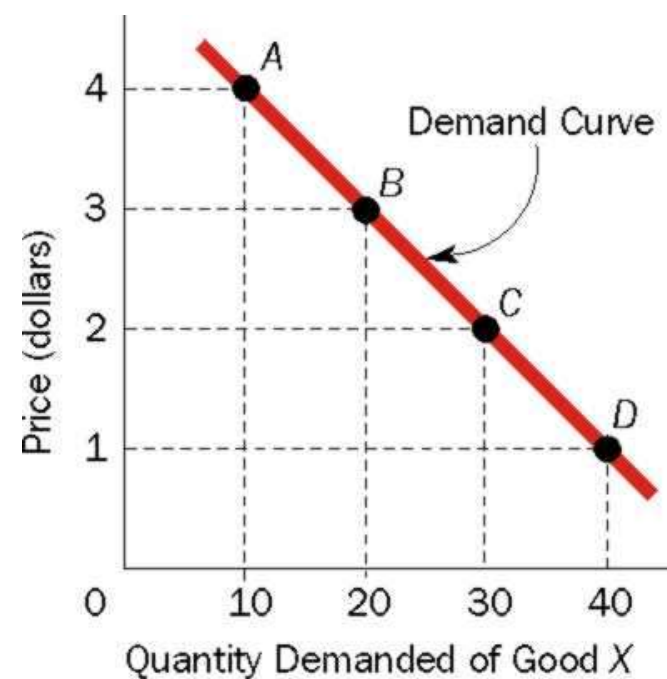
Quantity Going Down As Price Goes Up?

- People substitute lower-priced goods for higher-priced goods.
- The Law of Diminishing Utility: for a given time period, the marginal utility or satisfaction gained by consuming equal successive units of a good will decline as the amount consumed increases. According to the law, individuals obtain less utility from additional units of goods. So, they will only buy a larger amount of good at lower prices.
- Income effect: As price increases, real income falls.

Demand Schedule & The Demand Curve

Demand Schedule for Good X		
Price (dollars)	Quantity Demanded	Point in Part (b)
4	10	A
3	20	B
2	30	C
1	40	D

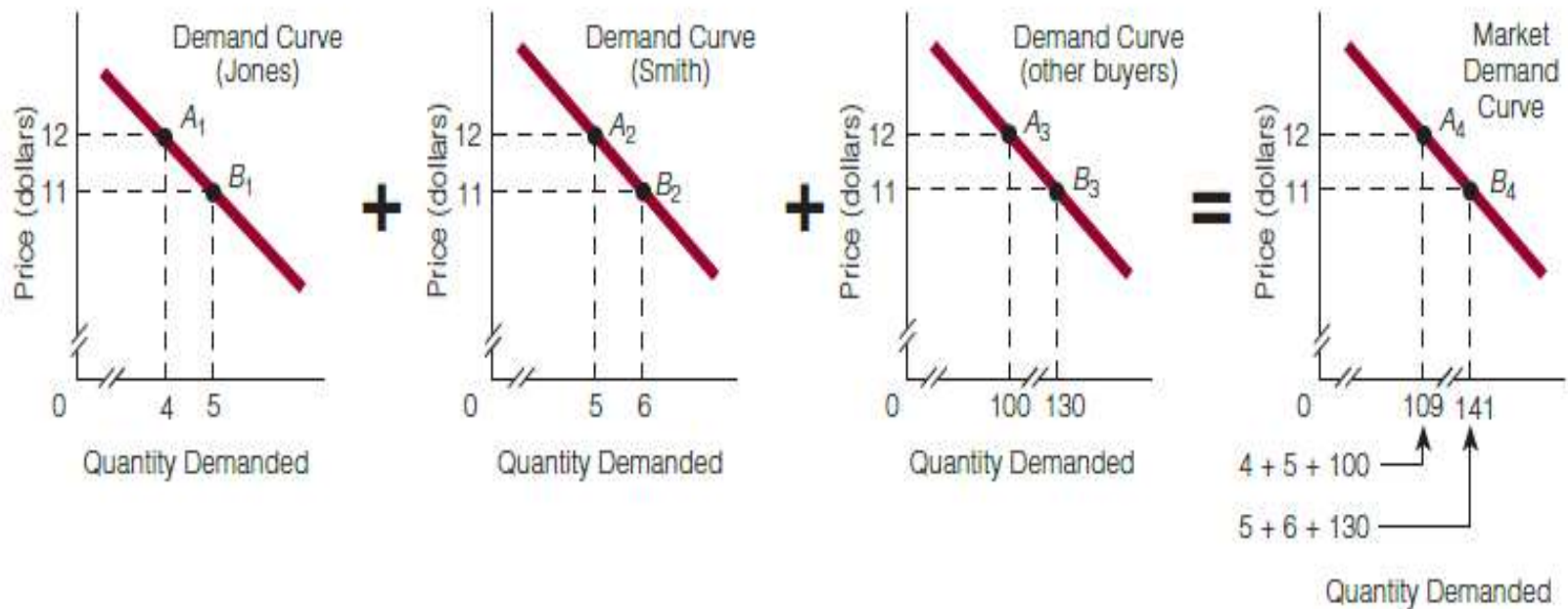
(a)



(b)

Market Demand Curve

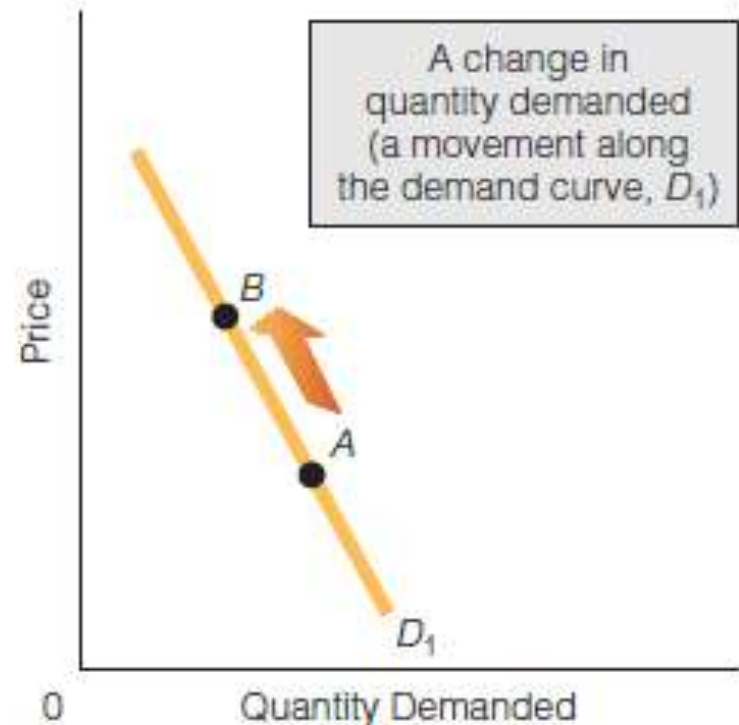
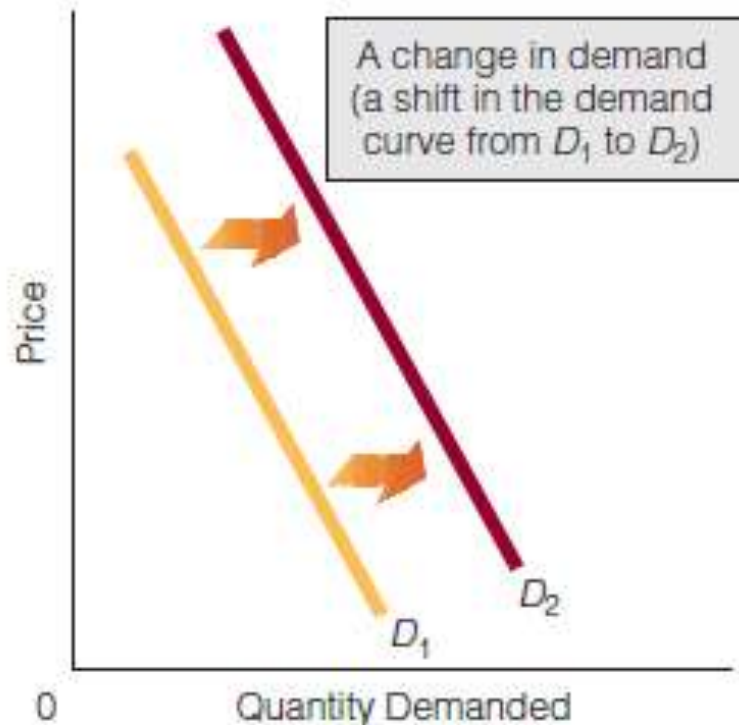
Horizontal summation of individual demand curves will give us the market demand curve.



A Change in Quantity Demanded Vs A Change in Demand

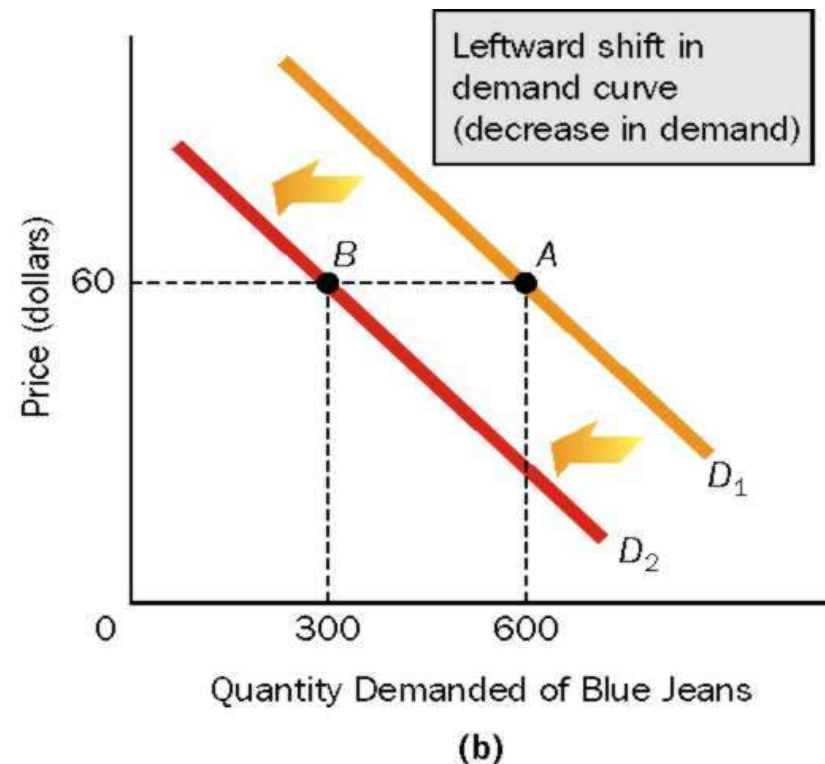
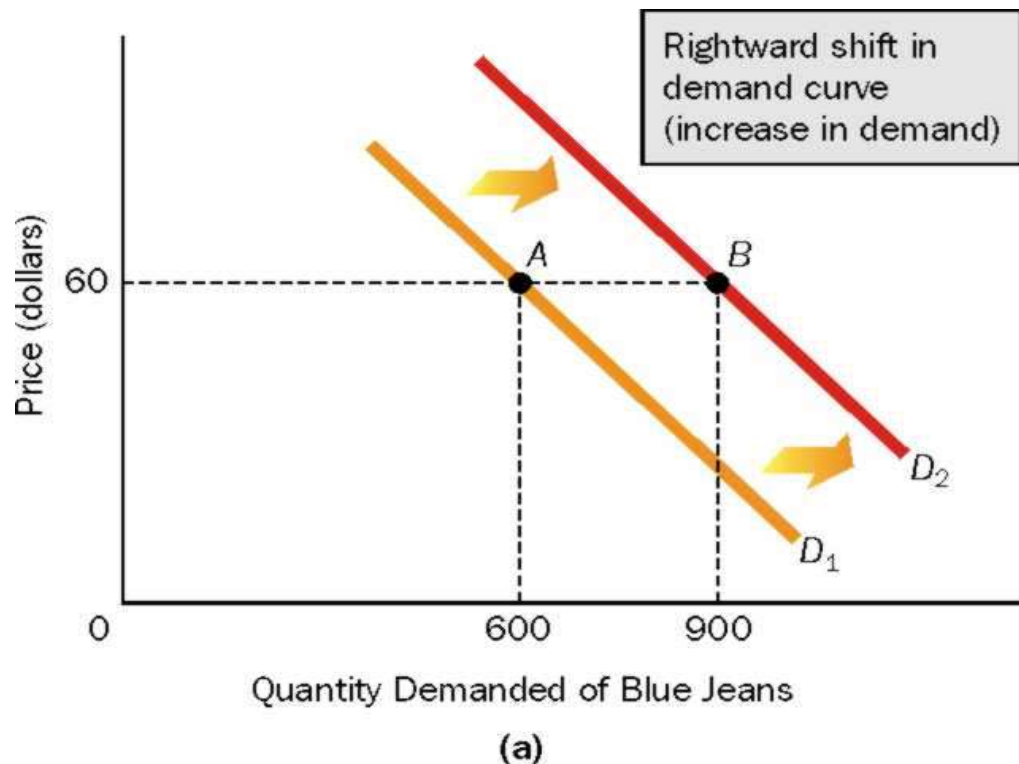
- Quantity demanded: The number of units of a good that individuals are willing and able to buy at a particular price.
- Change in quantity demanded: A movement from one point to another point on the same demand curve caused by a change in the price of the good.
- Change in demand = shift in demand curve.
- The demand for a good increases if people are willing and able to buy more of the good at all prices

Movement Vs shift



Shifting the Demand Curve

- A change in Demand causes a shift in the Demand curve.
- If Demand increases, the curve shifts to the right.
- If Demand decreases, the curve shifts to the left.



Shifts in Demand Curves

Shift Factors:

Income: If a person's income changes, his or her demand for a particular good may rise, fall, or remain constant.

- A normal good is a good the demand for which rises(falls) as income rises(falls).
- An inferior good is a good the demand for which rises(falls) as income falls(rises).
- For a neutral good, demand does not change as income rises or falls

Preferences: Preferences affect the amount of a good they are willing to buy at a particular price (Ex: favorite food, favorite author)

Prices of Related Goods:

If the demand for product X increases as the price for Y increases, and the demand for product X falls as the price for Y falls, X and Y are **substitutes** (Ex: Coke and Pepsi).

If the price of product A falls as the demand for product B rises, and the price of product A rises as the demand for product B falls, A and B are **complements** (Ex: Ketchup and Burger).

Number of Buyers: more buyers higher demand; fewer buyers, lower demand. The number of buyers in a particular market area may change due to change in birthrate, migration, immigration etc.

Expectations of Future Price: If people expect that price will increase in near future then people will buy the good now. So demand increases.

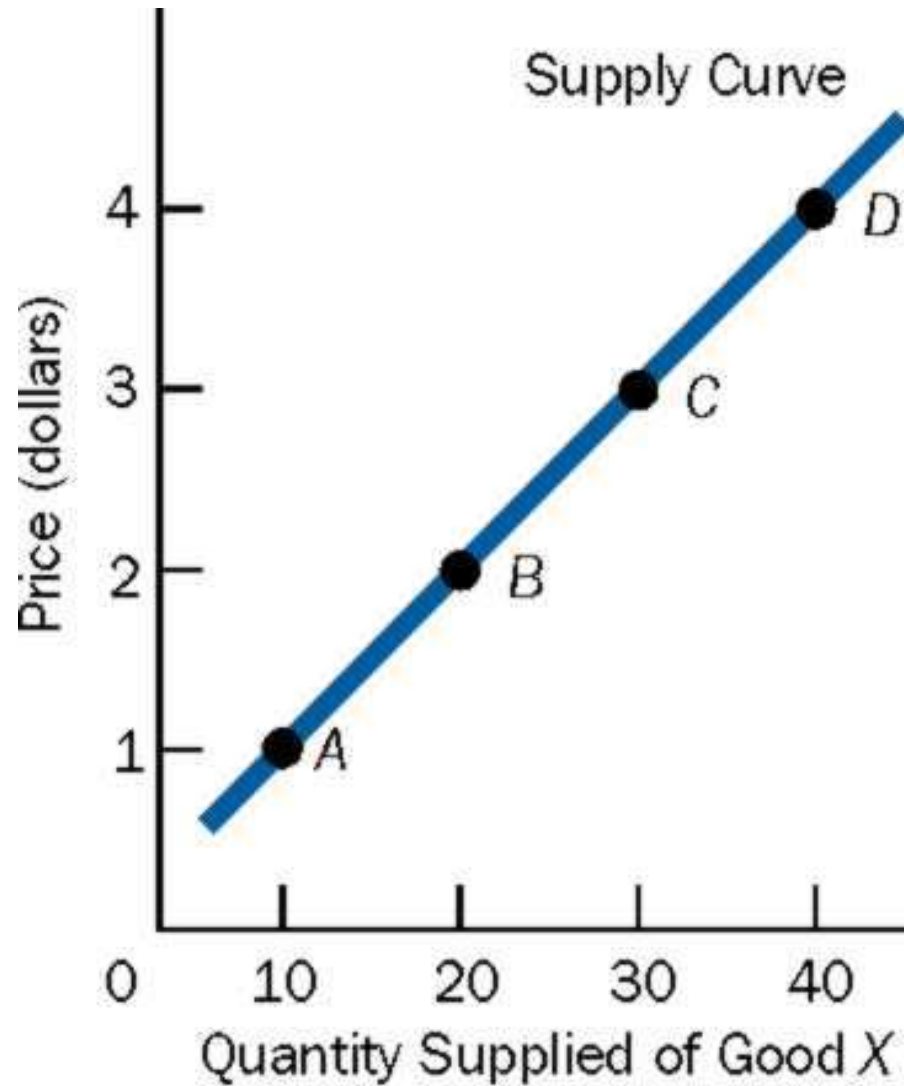
Q & A

- Why are demand curves downward sloping?
- Give an example that illustrates how to derive a market demand curve.
- Sandy plans to produce and sell flashlights and wants to know how many flashlights she will be able to sell. What would you tell her?

Supply

- Supply is the willingness and ability of sellers to produce and offer to sell different quantities of a good at different prices during a specific period of time
- Law of Supply: As the price of a good rises, the quantity supplied of the good rises.

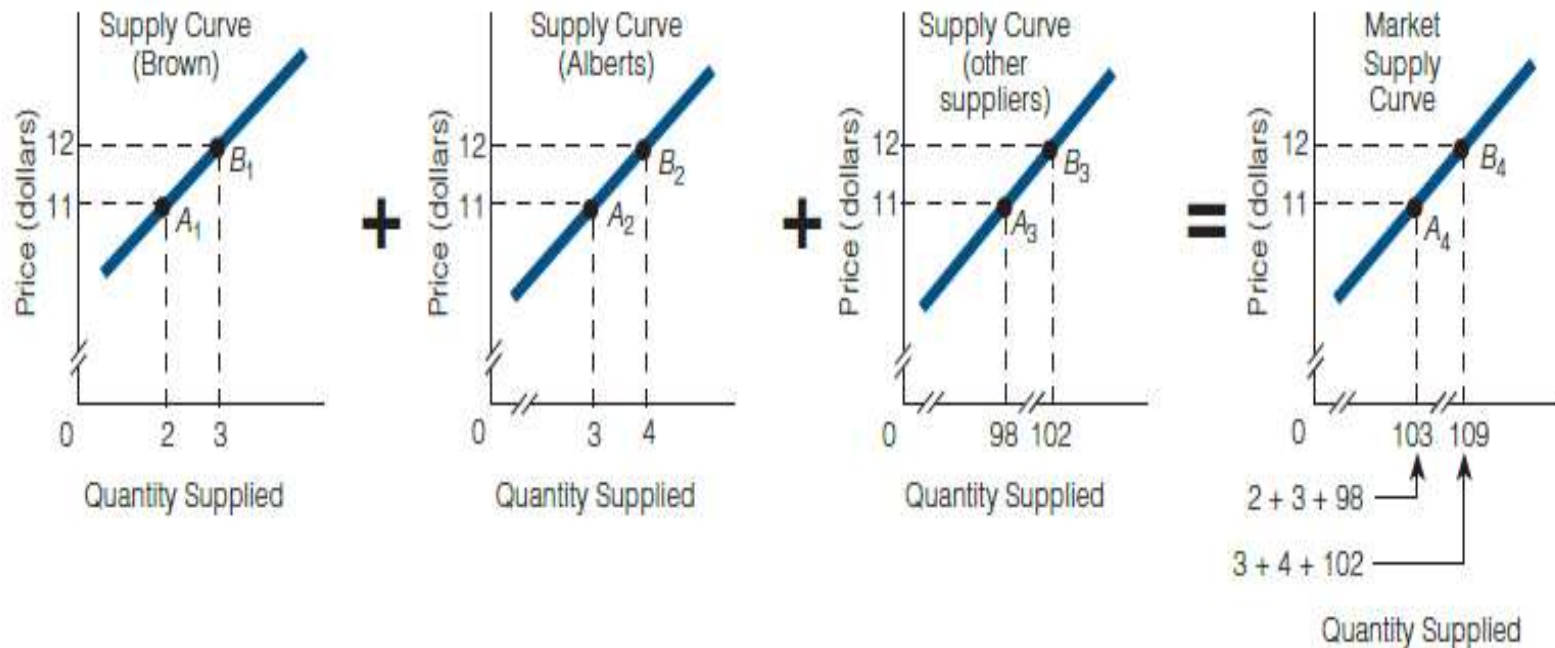
The Supply Curve



Slope of the Supply Curve:

- Increased production of a good comes at increased costs due to the law of diminishing marginal returns. An upward sloping supply curve simply reflects, under certain conditions, a higher price is an incentive to producers to produce more of the good. The incentive comes in the form of higher profits.
- Supply curve **is not upward** sloping when there is no time to produce additional units or when additional units cannot be produced.

Market Supply Curve: Adding individual supply curves gives us market supply curve.

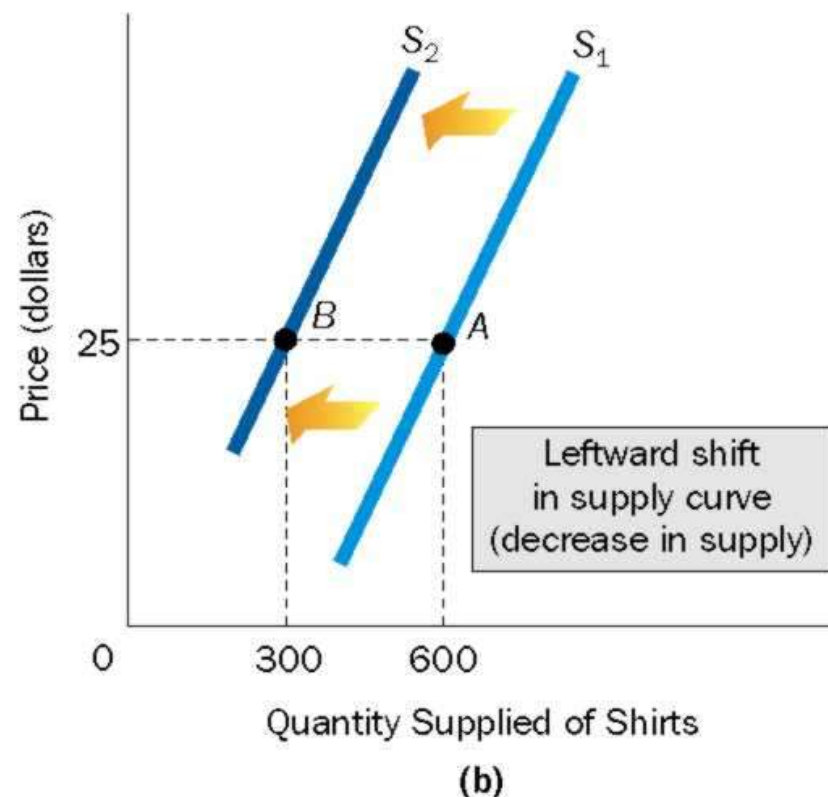
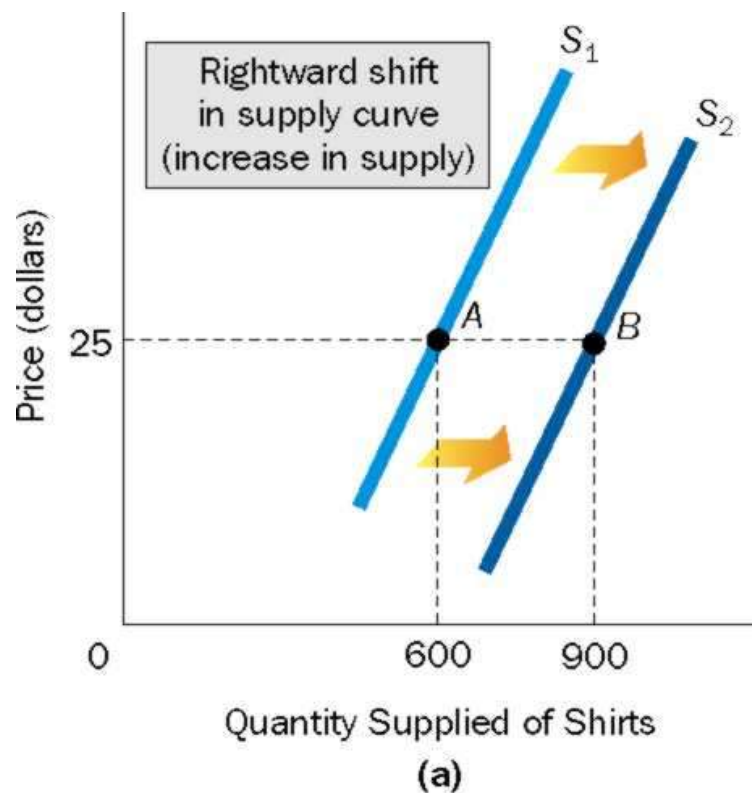


Shifting the Supply Curve

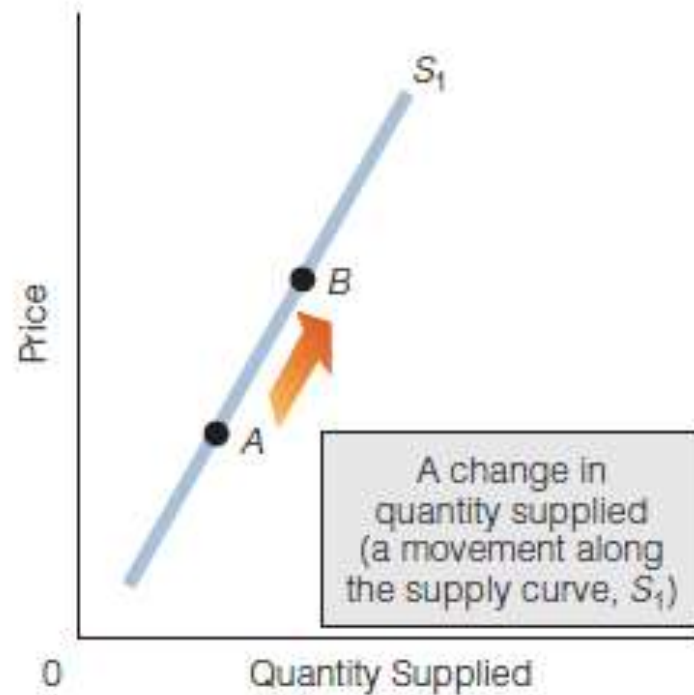
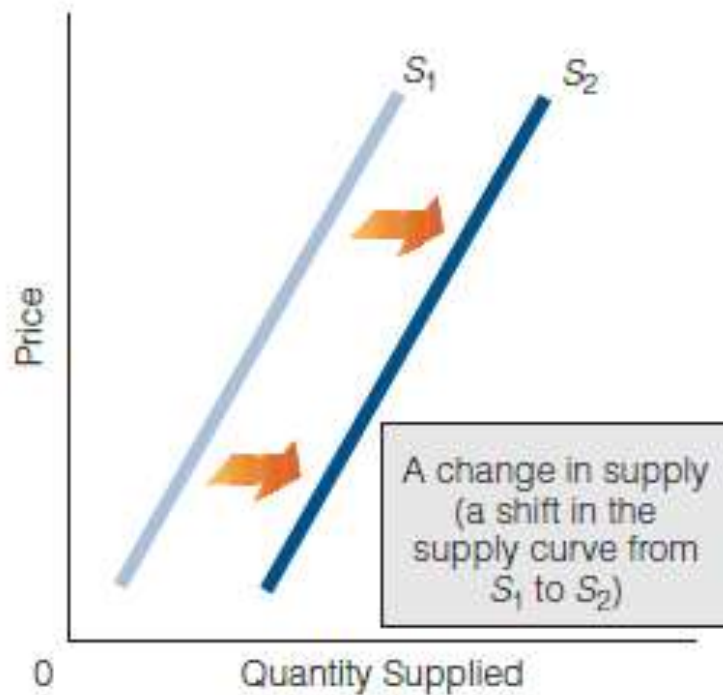
- Price of relevant resource: If the price of a relevant resource changes, the supply curve will shift (EX: wood prices increase, cost of a new house increases as well)
- Technology : Technology can increase the quantity supplied by producing more of a product with the same quantity of resources supplied.
- Price of other goods: Change in price of one good can led to a change in the supply of other good. (Ex: rice Vs jute production)
- If the **number of sellers** increase, the supply curve will shift.
- Expectations of future prices: If the price of a good is expected to be higher in the future, the supply curve will shift.
- Taxes and Subsidies: Taxes may increase per unit cost. Subsidy does the opposite.
- Government restrictions can change the supply curve by increasing or limiting production.

Shifting the Supply Curve

A Change in the Supply Curve is a shift in the Supply Curve, not merely moving up and down the same curve.



Movement Vs shift

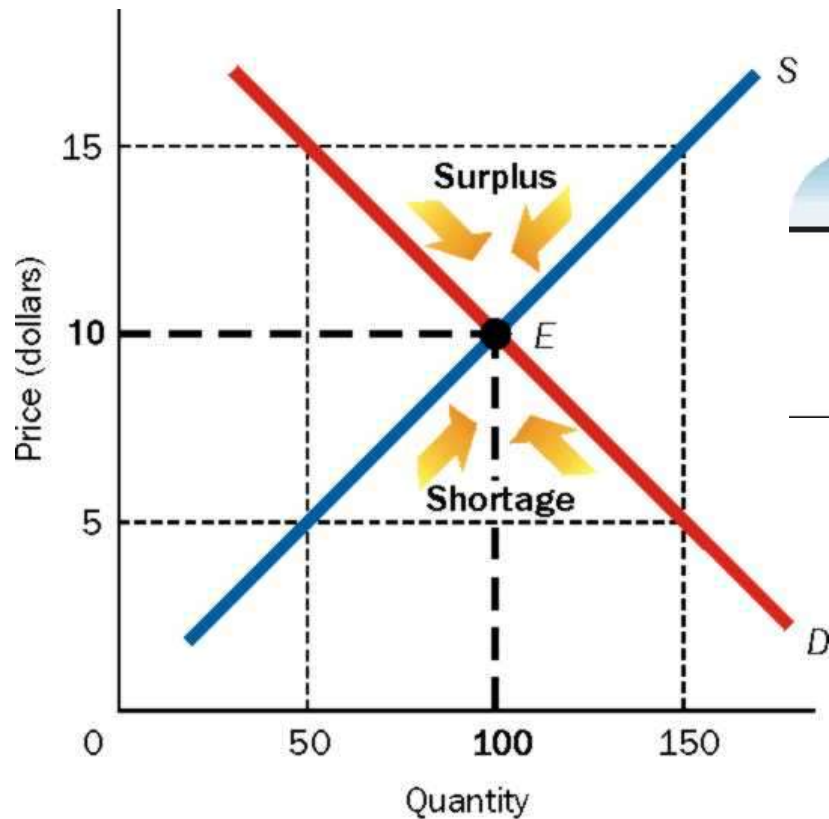


Q & A

- Which way (if any) does the Supply Curve shift if there is a decrease in the number of sellers? If there is a per-unit tax is placed on the production of the good? If the price of a relevant resource falls?

The Market

Putting Supply and Demand Together



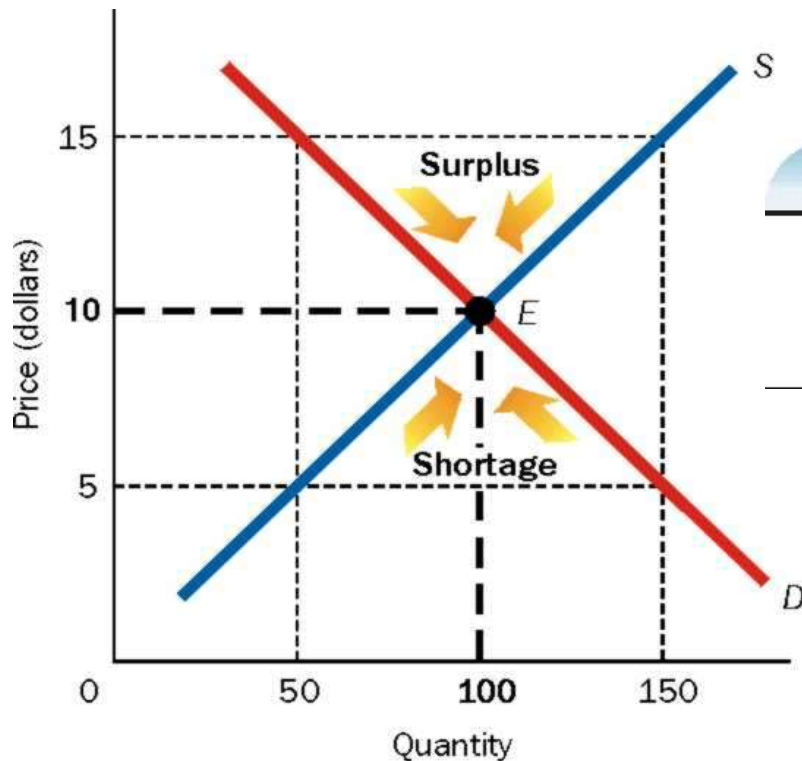
Price	Q_s	Q_d	Condition
\$15	150	50	Surplus
10	100	100	Equilibrium
5	50	150	Shortage

Market Language

- The price at which a quantity demanded equals the quantity supplied is the **equilibrium price**, or the market-clearing price.
- The quantity at which the amount of the good that buyers are willing and able to buy equals the amount that sellers are willing and able to sell and both equal the amount actually bought and sold is **the equilibrium quantity**.
- A condition at which quantity supplied is greater than the quantity demanded is called **surplus**.
- A condition at which quantity demanded is greater than the quantity supplied is called **shortage**.
- A market that has too much of a good or too little of a good is considered to be in disequilibrium.

Moving to Equilibrium

- Why does the price fall when there is a surplus?
- Why does the price rise when there is a shortage?
- Mutually beneficial exchange drives the market towards equilibrium.



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Equilibrium Price and Quantity Effects of Supply Curve Shifts and Demand Curve Shifts

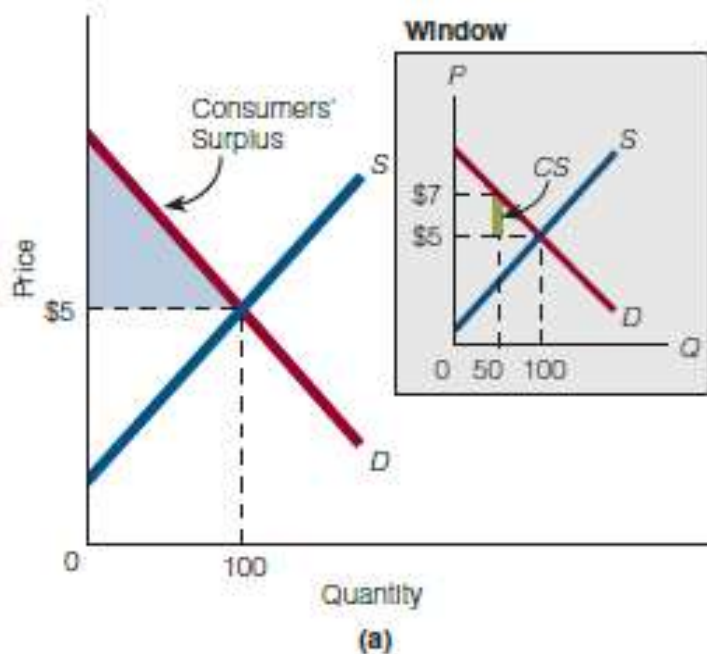
Equilibrium Price and Quantity Effects of Supply Curve Shifts and Demand Curve Shifts

Equilibrium in Terms of Consumers' and Producers' Surplus:

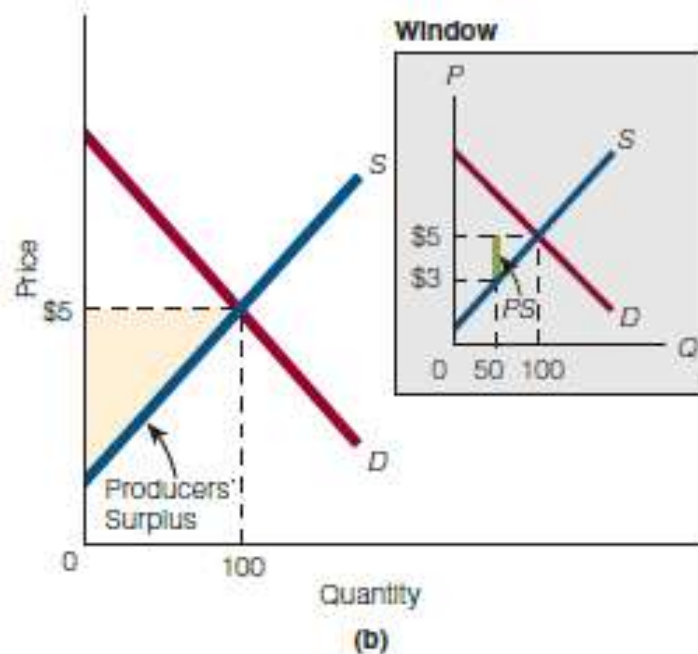
Consumers' surplus(CS) = Maximum buying price – Price paid

Producers' surplus(PS)= Price received – Minimum selling price

Total surplus = CS + PS

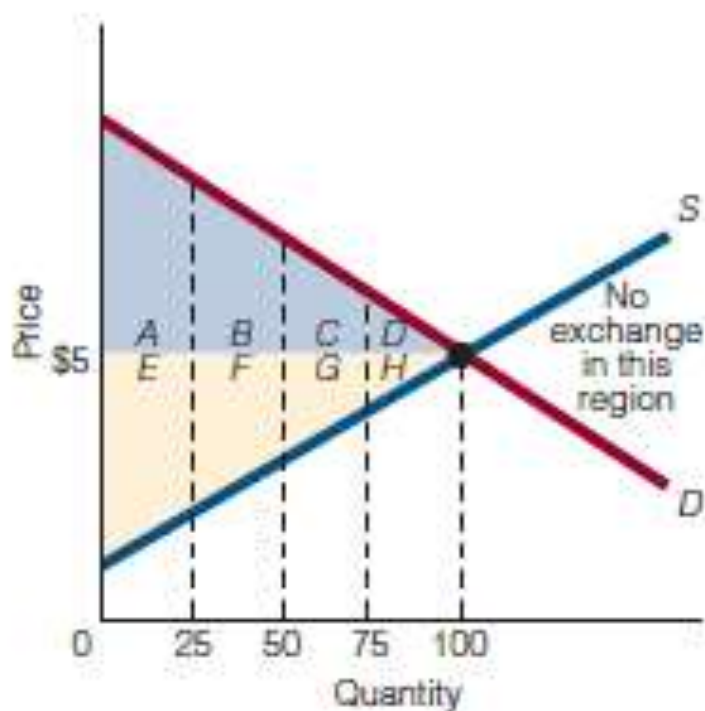


Consumers' Surplus (CS)



Producers' Surplus (PS)

At equilibrium, both consumers' surplus and producers' surplus are maximized or total surplus is maximized.



Price Controls

Price Ceiling: a government-mandated price above which legal trades may not be made.

Price Ceilings may cause:

- Shortages
- Fewer Exchanges
- Non-price Rationing Devices
- Buying and Selling at a prohibited Price
- Tie in Sales
- Buyers prefer a higher price
- Distort the flow of accurate information to the buyers

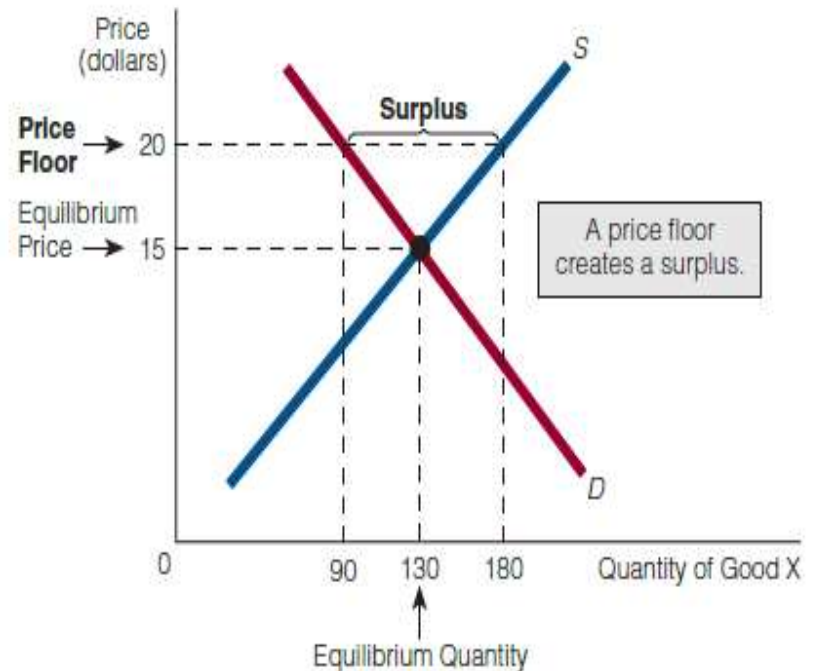


Price Floor

A price floor is a government mandated minimum price below which legal trades cannot be made.

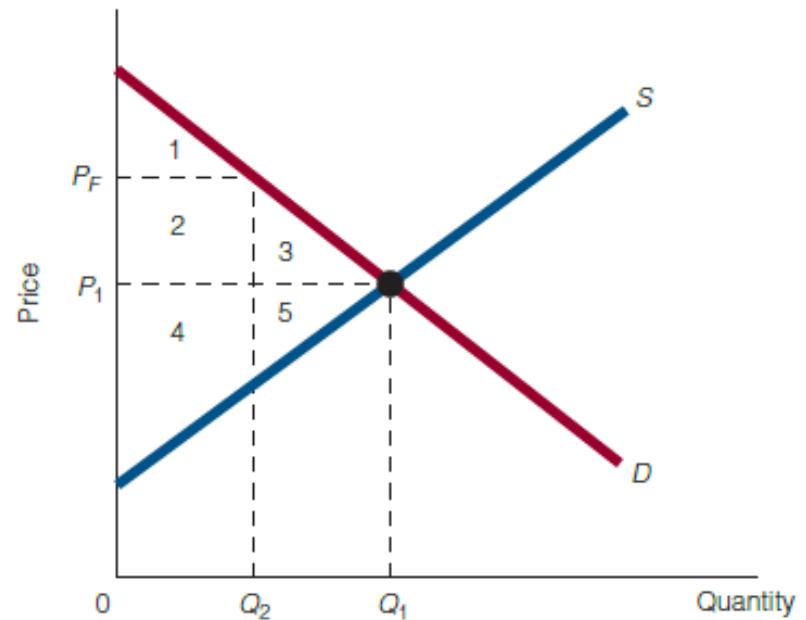
Price floors can cause-

- Surpluses
- Fewer Exchanges
- Deadweight loss



Deadweight Loss

Deadweight Loss The loss to society of not producing the competitive, or supply-and-demand determined, level of output.



Q & A

- Do buyers prefer lower prices to higher prices?
- Who might argue for a Price Ceiling? A Price Floor? Why would they argue their viewpoint?

